

SKILL WEEK 10

Task 1:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd = open("combined.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd < 0) return 1;

    dup2(fd, STDOUT_FILENO);
    dup2(STDOUT_FILENO, STDERR_FILENO);
    close(fd);

    printf("Standard output message.\n");
    fprintf(stderr, "Standard error message.\n");
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 10.1.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 10.1.c -o 10.1
ashishreddy@Ashishs-MacBook-Air ~ % ./10.1
ashishreddy@Ashishs-MacBook-Air ~ % cat combined.txt
Standard output message.
Standard error message.
```

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Task 2:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int p[2];
    pipe(p);

    if (fork() == 0) {
        dup2(p[1], STDOUT_FILENO);
        close(p[0]); close(p[1]);
        execlp("ls", "ls", NULL);
        return 1;
    }

    if (fork() == 0) {
        dup2(p[0], STDIN_FILENO);
        close(p[0]); close(p[1]);
        execlp("grep", "grep", ".c", NULL);
        return 1;
    }

    close(p[0]); close(p[1]);
    wait(NULL); wait(NULL);
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 10.2.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 10.2.c -o 10.2
ashishreddy@Ashishs-MacBook-Air ~ % ./10.2
10.1.c
10.2.c
9.1.c
9.2.c
9.3.c
9.4.c
```

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Task 3:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        for (int i = 1; i <= 5; i++) {
            printf("Background job: %d\n", i);
            sleep(1);
        }
        return 0;
    }

    printf("[1] Background job started with PID %d\n", pid);
    printf("Parent continues immediately.\n");
    waitpid(pid, NULL, 0);
    printf("Background job completed.\n");

    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 10.3.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 10.3.c -o 10.3
ashishreddy@Ashishs-MacBook-Air ~ % ./10.3
[1] Background job started with PID 4217
Parent continues immediately.
Background job: 1
Background job: 2
Background job: 3
Background job: 4
Background job: 5
Background job completed.
```

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Task 4:

Code:

```
#include <stdio.h>
#include <unistd.h>

typedef struct {
    int id;
    pid_t pid;
    char state[20];
} Job;

int main() {
    Job jobs[3] = {
        {1, 4217, "Running"},
        {2, 4218, "Stopped"},
        {3, 4219, "Running"}
    };

    printf("JOB TABLE\n");
    printf("ID\tPID\tSTATE\n");

    for (int i = 0; i < 3; i++)
        printf("%d\t%d\t%s\n",
            jobs[i].id, jobs[i].pid, jobs[i].state);

    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 10.4.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 10.4.c -o 10.4
ashishreddy@Ashishs-MacBook-Air ~ % ./10.4
JOB TABLE
ID PID STATE
1 4217 Running
2 4218 Stopped
3 4219 Running
```